

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – XI

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Practical 11: Optimization Modeling and Simulation Modeling

Part A: Optimization Modeling

Business Problem: A company produces two products, Product A and Product B. The company wants to maximize its profit using the available machine and labour hours.

Given data:

Product Profit per Unit Machine Hours Labour Hours

A 40 2 1

B 30 1 2

Available Machine Hours = 100

Available Labour Hours = 80

Practical Questions:

1. Define the decision variables for Product A and Product B.
2. Define the objective of the problem.
3. Formulate the objective function to maximize the total profit.
4. Formulate the machine-hour constraint.
5. Formulate the labour-hour constraint.
6. Add non-negative constraints for the decision variables.
7. Solve the linear programming problem using Python.
8. Find the optimal number of units of Product A and Product B.
9. Calculate the maximum profit.
10. Check whether the available machine and labour hours are satisfied.
11. Display the optimal solution in a suitable format.
12. Interpret the optimal solution in a business context.
13. Write a conclusion based on the optimization result.

CODE:

```
# Practical 11: Optimization Modeling
# Part A: Optimization Modeling
from scipy.optimize import linprog
# -----
# Decision Variables
# -----
# x = Number of units of Product A
# y = Number of units of Product B
# -----
# Objective Function
# Maximize:
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# Profit = 40x + 30y
#
# scipy.optimize.linprog performs minimization,
# so we use negative profit values.
# -----
profit = [-40, -30]
# -----
# Constraints
# -----
# Machine Hours:
# 2x + y <= 100
# Labour Hours:
# x + 2y <= 80
A = [
    [2, 1],
    [1, 2]
]
b = [100, 80]
# -----
# Non-negative Constraints
# x >= 0
# y >= 0
# -----
bounds = [
    (0, None),
    (0, None)
]
# -----
# Solve Linear Programming Problem
# -----
result = linprog(
    profit,
    A_ub=A,
    b_ub=b,
    bounds=bounds,
    method="highs"
)
# -----
# Display Result
# -----
if result.success:
    product_A = result.x[0]
    product_B = result.x[1]
    maximum_profit = -result.fun
    # Calculate resource usage
    machine_used = (2 * product_A) + product_B
    labour_used = product_A + (2 * product_B)
    print("\n=====")
    print("    OPTIMIZATION MODELING")

```

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print("=====")
print("\nOptimal Production:")
print("-----")
print(f"Product A Units : {product_A:.0f}")
print(f"Product B Units : {product_B:.0f}")
print("\nProfit:")
print("-----")
print(f"Maximum Profit : ₹{maximum_profit:.2f}")
print("\nResource Usage:")
print("-----")
print(f"Machine Hours Used : {machine_used:.0f} / 100")
print(f"Labour Hours Used : {labour_used:.0f} / 80")
print("\nConstraint Check:")
print("-----")
if machine_used <= 100:
    print("Machine Hours Constraint: Satisfied")
else:
    print("Machine Hours Constraint: Not Satisfied")
if labour_used <= 80:
    print("Labour Hours Constraint : Satisfied")
else:
    print("Labour Hours Constraint : Not Satisfied")
else:
    print("No optimal solution found.")

```

OUTPUT:

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=====
      OPTIMIZATION MODELING
=====

Optimal Production:
-----
Product A Units : 40
Product B Units : 20

Profit:
-----
Maximum Profit   : ₹2200.00

Resource Usage:
-----
Machine Hours Used : 100 / 100
Labour Hours Used  : 80 / 80

Constraint Check:
-----
Machine Hours Constraint: Satisfied
Labour Hours Constraint : Satisfied

```

Part B: Simulation Modeling

Business Problem: A supermarket wants to decide the appropriate inventory level for a product when daily demand is uncertain.

Practical Questions:

14. Generate random daily demand values for a product for 30 days.
15. Display the generated demand values.
16. Select an inventory level of 30 units.
17. Compare the daily demand with the available inventory.
18. Calculate the number of units sold each day.
19. Calculate the number of units left after each day.
20. Calculate the shortage when demand is greater than available inventory.
21. Calculate the total units sold and total shortage for 30 days.
22. Repeat the simulation using inventory levels of 30, 40, and 50 units.
23. Store the results of all three scenarios in a DataFrame.
24. Compare the total sales and total shortage for each inventory level.
25. Create a chart to compare the results of the three scenarios.
26. Identify how changing the inventory level affects sales and shortage.
27. Write a conclusion about the inventory level based on the simulation results.

CODE:

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# Practical 11: Simulation Modeling
# Part B: Supermarket Inventory Simulation
import pandas as pd
import matplotlib.pyplot as plt
# -----
# 1. Load Kaggle Dataset
# -----
file_name = "retail_store_inventory.csv"
df = pd.read_csv(file_name)
print("\n=====")
print("    DATASET INFORMATION")
print("=====")
print("\nFirst 5 Records:")
print(df.head())
print("\nDataset Columns:")
print(df.columns.tolist())
# -----
# 2. Generate 30 Daily Demand Values
# -----

# Select first 30 demand values from dataset
demand = df["Demand"].head(30).tolist()
days = list(range(1, 31))
print("\n=====")
print("    DAILY DEMAND - 30 DAYS")
print("=====")
for day, d in zip(days, demand):
    print(f"Day {day}: {d} units")
```

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# -----
# 3. Inventory Simulation Function
# -----
def simulate_inventory(demand, inventory_level):
    units_sold = []
    units_left = []
    shortage = []
    for daily_demand in demand:
        # Units sold cannot be greater than inventory
        sold = min(daily_demand, inventory_level)
        # Units remaining
        remaining = max(inventory_level - daily_demand, 0)
        # Shortage
        short = max(daily_demand - inventory_level, 0)

        units_sold.append(sold)
        units_left.append(remaining)
        shortage.append(short)
    return units_sold, units_left, shortage
# -----
# 4. Simulation for Inventory = 30
# -----
sales_30, left_30, shortage_30 = simulate_inventory(
    demand,
    30
)
simulation_30 = pd.DataFrame({
    "Day": days,
    "Demand": demand,
    "Inventory": 30,
    "Units Sold": sales_30,
    "Units Left": left_30,
    "Shortage": shortage_30
})
print("\n=====")
print("  SIMULATION - INVENTORY = 30")
print("=====")
print(simulation_30)
print("\nTotal Units Sold:",
      simulation_30["Units Sold"].sum())
print("Total Shortage:",
      simulation_30["Shortage"].sum())
# -----
# 5. Simulate Inventory Levels
# 30, 40 and 50
# -----
inventory_levels = [30, 40, 50]
results = []
for inventory in inventory_levels:

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sales, left, shortage = simulate_inventory(
    demand,
    inventory
)
total_sales = sum(sales)
total_shortage = sum(shortage)
results.append({
    "Inventory Level": inventory,
    "Total Sales": total_sales,
    "Total Shortage": total_shortage
})
# -----
# 6. Store Results in DataFrame
# -----
results_df = pd.DataFrame(results)
print("\n=====")
print("    INVENTORY COMPARISON")
print("=====")
print(results_df)
# -----
# 7. Display Best Sales and Shortage Results
# -----
print("\n=====")
print("    ANALYSIS")
print("=====")
highest_sales = results_df.loc[
    results_df["Total Sales"].idxmax()
]
lowest_shortage = results_df.loc[
    results_df["Total Shortage"].idxmin()
]
print(
    f"Highest Total Sales: "
    f"Inventory {highest_sales['Inventory Level']} units"
)
print(
    f"Lowest Total Shortage: "
    f"Inventory {lowest_shortage['Inventory Level']} units"
)
# -----
# 8. Create Chart
# -----
plt.figure(figsize=(9, 6))
plt.plot(
    results_df["Inventory Level"],
    results_df["Total Sales"],
    marker="o",
    label="Total Sales"
)

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plt.plot(
    results_df["Inventory Level"],
    results_df["Total Shortage"],
    marker="o",
    label="Total Shortage"
)
plt.xlabel("Inventory Level")
plt.ylabel("Units")
plt.title(
    "Inventory Level vs Total Sales and Total Shortage"
)
plt.legend()
plt.grid(True)
plt.show()
# -----
# 9. Save Results
# -----
results_df.to_csv(
    "inventory_simulation_results.csv",
    index=False
)

```

OUTPUT:

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DATASET INFORMATION
-----

First 5 Records:
  Date Store ID Product ID Category Region Inventory Level \
0 2022-01-01  S001      P0001 Electronics North         195
1 2022-01-01  S001      P0002 Clothing North         117
2 2022-01-01  S001      P0003 Clothing North         247
3 2022-01-01  S001      P0004 Electronics North         139
4 2022-01-01  S001      P0005 Groceries North         152

  Units Sold  Units Ordered  Price  Discount Weather Condition Promotion \
0         102           252  72.72         5      Snowy          0
1         117           249  80.16        15      Snowy          1
2         114           612  62.94        10      Snowy          1
3          45           102  87.63        10      Snowy          0
4          65           271  54.41         0      Snowy          0

  Competitor Pricing Seasonality Epidemic Demand
0          85.73      Winter         0      115
1          92.02      Winter         0      229
2          60.08      Winter         0      157
3          85.19      Winter         0       52
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ANALYSIS
-----
Highest Total Sales: Inventory 50 units
Lowest Total Shortage: Inventory 50 units
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